

Homework 4 (100pt)

1. Given a random key K of length m bits. K is used as a one time-pad to encrypt plaintext message $P1$ (of length m bits) to obtain ciphertext $C1$. K is then also used to encrypt plaintext $P2$ (of length m bits) to obtain ciphertext $C2$ (i.e., the key is reused!). If the adversary is able to obtain $C1$, $C2$, and $P2$ (known plaintext attack), explain how he can compute $P1$. (10pts)
2. Alice wants to send some messages to Bob. Both Alice and Bob are aware that the communication channel between them is insecure. An adversary may modify the messages or impersonate Alice to Bob. They want to ensure the integrity of the messages. Answering the following questions.
 - a. How can they detect unauthorized modifications of messages only with secret key cryptography? Assume they have a way to share a secret key. (3pts)
 - b. How can they achieve this goal with hash function? Assume they have a way to share a secret key. (3pts)
 - c. How can they achieve this goal with public key cryptography. Assume they both know the certificate of Alice. (4pts)

3. Programmer Joe is told to design a scheme to prevent messages from being modified by an attacker. Joe decides to append to each message a hash of that message. Why doesn't this solve the problem? (10pts)
4. What are the desired properties of hash functions? (10pts)
5. Hash functions are reasonably fast, but we can create a faster hash function. Take your message, divide it into 128-bit chunks, and then XOR the chunks together to get a 128-bit result. Apply the standard message digest (e.g., MD5 or SHA1) on the result. Is this a good hash function? Why or why not? (10pts)
6. For a Diffie-Hellman key exchange, $p = 7$ and $g = 3$. If Alice sends to Bob the value 6 and Bob sends to Alice the value 5, what is Alice's private key S_A , what is Bob's private key S_B , and what is the shared secret key on which they agree? (10pts)
7. RSA algorithms are widely used to generate public and private key pairs. Let e and d denote the public and private key respectively. Answer the following questions:
 - a. A public key is used to encrypt a plaintext M ($M < n$). What is the ciphertext? (10pts)
 - b. A private key is used to decrypt a ciphertext C . What is the decryption result? (10pts)
 - c. A private key is used to sign a plaintext M ($M < n$). What is the signature? (10pts)
 - d. Alice sends a message M and the message's signature S to Bob. How can Bob verify that the received M and S are really generated by Alice? Assume that Bob knows Alice's public key. (10pts)